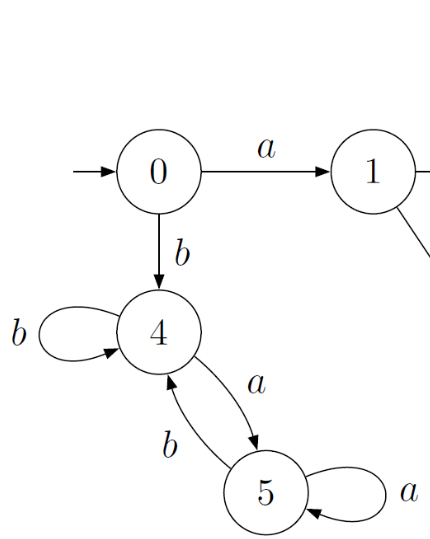


G52LAC (COMP2040)
Languages and Computation

Lecture 2: Alphabets, Words and Languages

Tomas Maul
School of Computer Science
University of Nottingham Malaysia Campus

Outline

- Quick recap. of the previous lecture.
- Quick revision: sets, ordered tuples, functions and relations.
- Terminology.
- Symbols & alphabets.
- Words.
- Languages.
- Revisiting the Chomsky Hierarchy.

Sets vs. ordered n-tuples

- Ordered n-tuples vs. sets.
- Ordered n-tuples:
  - Order matters.
  - Elements can repeat.
  - E.g. (3,2,2,5,9,5).
- Sets:
  - Order does not matter.
  - Elements can't repeat.
  - E.g. {2,3,5,9}.

Relation?
Function?
Comparison?

Functions and Relations

- Cartesian product between two sets A and B:
  - The set of all ordered 2-tuples (a,b) where a belongs to A and b belongs to B.
  - Whiteboard: A = {1,2} and B = {5,6,7} …
- A relation is a generalization of a function.
- Consider two sets A and B.
  - A x B. Cartesian product between A and B.
  - Any subset of A x B is a relation between A and B.
  - A function is a special relation R between A and B:
    - Whenever (a,b) and (a,b’) are in the relation R, then b = b’.

Terminology

- The terms alphabet, word and language are used in a strict technical sense in this course.
- An alphabet is a finite set of symbols.
- A word is a finite sequence of symbols.
- A language is a set of words.
- Languages can be finite or infinite.
- The term string is often used interchangeably with the term word.

Symbols & Alphabets

- What is a symbol, then?
- Anything, but it has to come from an alphabet.
- Usually, Σ is used to denote an alphabet.
- Example alphabets:

$$\begin{aligned}
\Sigma_1 &= \{0, 1\} \\
\Sigma_2 &= \{a, b, c, d, e, f, g, h, i, j, k, l, m, \\
&\quad n, o, p, q, r, s, t, u, v, w, x, y, z\} \\
\Sigma_3 &= \{\circ, \square, \triangle\} \\
\Sigma_4 &= \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, +, -, *, /\}
\end{aligned}$$

- Important exception: ε is never used as an alphabet symbol.

The Empty Word

- ε is used to denote the empty word: the sequence of zero symbols.
- But ε itself is not a symbol!
- ε is a word, not a set.
- So don't confuse it with the empty set (denoted $\emptyset$ or $\{\,\}$).
- Thus, $\{\varepsilon\} \neq \{\,\}$.

Words over an alphabet

- The set of all words over an alphabet Σ is denoted by Σ^*.
- Σ^* can be defined inductively as follows:
  - $\varepsilon \in \Sigma^*$
  - if $x \in \Sigma$ and $w \in \Sigma^*$ then $xw \in \Sigma^*$

Inductive or recursive definition

- Note that $\varepsilon \in \Sigma^*$ for any alphabet Σ (including $\Sigma = \emptyset$).
- Iff $\Sigma \neq \emptyset$ then Σ^* is an infinite set (of finite words).

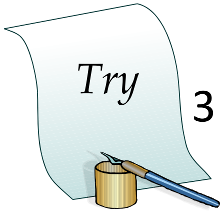
So when can Σ^* be finite?

Note 1: a word is essentially an ordered n-tuple without punctuation, e.g. the word *cat* corresponds to the ordered 3-tuple (c,a,t).

Note 2: The inductive definition of a set S consists of first specifying one or more elements of the set S and then specifying one or more construction rules to obtain other elements of S based on existing elements of S.

Example

- Given $\Sigma = \{0, 1\}$, some elements of Σ^* are:



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Recall:

- The set of all words over an alphabet Σ is denoted by Σ^*.
- Σ^* can be defined inductively as follows:
  - $\varepsilon \in \Sigma^*$
  - if $x \in \Sigma$ and $w \in \Sigma^*$ then $xw \in \Sigma^*$
- Note that $\varepsilon \in \Sigma^*$ for any alphabet Σ (including $\Sigma = \emptyset$).
- Iff $\Sigma \neq \emptyset$ then Σ^* is an infinite set (of finite words).

Example

- Given $\Sigma = \{0, 1\}$, some elements of Σ^* are:

  ε,
  $0, 1$,
  $\mathit{00}, 10, \mathit{01}, \mathit{11}$,
  $\mathit{000}, \mathit{100}, \mathit{010}, \mathit{110}, \mathit{001}, \mathit{101}, \mathit{011}, \mathit{111}$,
  $\mathit{0000}, \ldots$

- This is just applying the inductive definition.
- Important note: only write ε if it appears on its own, as it denotes an absence of symbols.

  e.g. *000ε* is redundant; you can just depict the word as *000*.

Alternative Notation

- The set of all words over Σ of length n is denoted by Σ^n (where $n \in \mathbb{N}$).
- For example, if $\Sigma = \{a, b\}$, then $\Sigma^2 = \{aa, ab, ba, bb\}$.
- This can be used to give an alternative (but equivalent) definition of Σ^*:

$$\Sigma^* = \bigcup_{n=0}^{\infty} \Sigma^n$$

- Remember that in computer science, $0 \in \mathbb{N}$.

Languages

- A language L over an alphabet Σ is a subset of Σ^*:

  $$L \subseteq \Sigma^*$$

  or

  $$L \in \mathcal{P}\left(\Sigma^*\right)$$

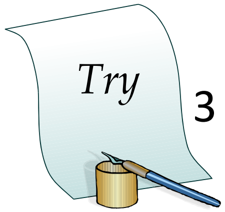
Remember “power set”?

- A language may be a finite or infinite set.
- Note that while ε is always an element of Σ^*, it may or may not be an element of an arbitrary language.

Exercise

Given $\Sigma = \{a, b, c\}$, define some languages over Σ.

Try to define some finite ones and some infinite ones.



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Exercise

Given $\Sigma = \{a, b, c\}$, define some languages over Σ.

- $\{a, abba, baa, cab\}$
- $\{c\}$
- $\{\varepsilon, a, bbb\}$
- $\{\varepsilon\}$
- $\{a^n \mid n \in \mathbb{N}\}$
- $\{a^n b^n \mid n \in \mathbb{N}, n \geqslant 10\}$
- $\{w \mid w \in \Sigma^*, odd\,(length\,(w))\}$
- $\emptyset$
- Σ^*

Concatenation of Words

- An important operation on words is concatenation.
- Concatenation is denoted by juxtaposition (i.e. writing the words side by side without using an operator symbol).
- If $v \in \Sigma^*$ and $w \in \Sigma^*$ then $vw \in \Sigma^*$
- Concatenation can be defined by primitive recursion:

$$\begin{aligned} \varepsilon w &= w \\ (xv)\, w &= x\, (vw) \end{aligned}$$

where

$$\begin{aligned} x &\in \Sigma \\ v, w &\in \Sigma^* \end{aligned}$$

Properties of Word Concatenation

- Concatenation is associative and has unit ε:

$$
\begin{aligned}
u\,(vw) &= (uv)\,w \\
\varepsilon u = u &= u\varepsilon
\end{aligned}
$$

  where

$$u, v, w \in \Sigma^*$$

- Concatenation of words is not commutative (i.e. order matters), as words are sequences.

$$vw \neq wv$$

Concatenation of Languages

- Remember, languages are sets, not sequences.
- Given two languages M and N over an alphabet Σ, their concatenation (MN) is defined:

$$MN = \{uv \mid u \in M \wedge v \in N\}$$

- Example:

$$\begin{aligned}
\Sigma &= \{a, b, c\} \\
M &= \{\varepsilon, a, aa\} \\
N &= \{b, c\} \\
MN &= \{uv \mid u \in \{\varepsilon, a, aa\} \wedge v \in \{b, c\}\} \\
&= \{\varepsilon b, \varepsilon c, ab, ac, aab, aac\} \\
&= \{b, c, ab, ac, aab, aac\}
\end{aligned}$$

Properties of Language Concatenation (1)

- Concatenation of languages is associative:

$$L\,(MN) \;=\; (LM)\,N$$

- Concatenation of languages has zero $\emptyset$ (the empty language):

$$L\emptyset \;=\; \emptyset \;=\; \emptyset L$$

- Concatenation of languages has unit $\{\varepsilon\}$ (the language containing only the empty word):

$$L\,\{\varepsilon\} \;=\; L \;=\; \{\varepsilon\}\,L$$

Properties of Language Concatenation (2)

- Concatenation of languages distributes through set union:

$$\begin{aligned} L\,(M \cup N) &= LM \cup LN \\ (L \cup M)\,N &= LN \cup MN \end{aligned}$$

- But it does not distribute through set intersection:

$$L\,(M \cap N) \neq LM \cap LN$$

- Counterexample:

$$\begin{aligned} L &= \{\varepsilon, a\}, M = \{\varepsilon\}, N = \{a\} \\ L\,(M \cap N) &= L\emptyset = \emptyset \\ LM \cap LN &= \{\varepsilon, a\} \cap \{a, aa\} = \{a\} \end{aligned}$$

Concatenating a Language with Itself

- A language can be concatenated with itself.
- Exponent notation is often used for this:
  - $L^1 = L$
  - $L^2 = LL$
  - $L^3 = LLL$
  - $L^4 = LLLL$
  - etc. . .
- L^0 is defined to be $\{\varepsilon\}$.
  (As $\{\varepsilon\}$ is the unit of concatenation.)

Kleene Star

- Given $L \subseteq \Sigma^*$, L^* is zero or more concatenations of L.
- Note that these are different stars (but both mean 'zero or more').

$$L^* = \{w_0 w_1 \dots w_{n-1} \mid n, i \in \mathbb{N}, \forall i < n, w_i \in L\}$$

or

$$L^* = \bigcup_{n=0}^{\infty} L^n = L^0 \cup L^1 \cup L^2 \cup \dots$$

or

$$\begin{array}{rcl} & & \varepsilon \in L^* \\ w \in L & \Rightarrow & w \in L^* \\ v \in L^* \wedge w \in L^* & \Rightarrow & vw \in L^* \end{array}$$

Language Membership

- Fundamental question of this module:

  Given a language $L \subseteq \Sigma^*$ and a word $w \in \Sigma^*$, can we determine if $w \in L$?

- If L is finite, this is easy.
- But not so easy if L is infinite, which most interesting languages are.
- We need:
  - A finite (and preferably concise) description of the (infinite) language.
  - A method to decide if $w \in L$ or not, given such a description.
- Over the course of this module we are going to encounter a number of possibilities, with varying descriptive power.

Recommended Reading

- Introduction to Automata Theory, Languages, and Computation (3rd edition), pages 28–33.
- Original G52MAL Lecture Notes, page 6.

Easy Exercises

- An alphabet is a _finite_ _set_ of symbols.
- A word is a ______ _______ of symbols.
- A language is a _______ of words.
- Can languages be infinite?

Easy Exercises

- What is ε?
- Is it a set? Is it the same as { } ?

Easy Exercises

- Given an alphabet Σ, explain in your own words the meaning of Σ^*.
- Given $\Sigma = \{a,b,c\}$, list five elements from Σ^*.

Easy Exercises

- Consider the languages L1 = {1,22} and L2 = {a,bb}.
- List down the elements of L3, which is defined as the concatenation L1 L2.

Question from a previous exam

(a) [A] Consider the following languages: **[4 marks]**

L1 = {a,b,c}
L2 = {c,d,e}
L3 = {c,d,f}
L4 = {g,h}

List all words belonging to the following languages:
[1 mark each]
L5 = $(L1 \cup L2) \cap L3$
L6 = $(L1 \cap L2)(L2 \cap L3)$
L7 = $(\{\varepsilon\}L1\{\varepsilon\}) \cap L3$
L8 = $(L1 \cup L3)(L3 \cap L4)$

Question from a previous exam

(a) [A] Consider the following languages: **[4 marks]**

$L1$ = {a,b,c}
$L2$ = {c,d,e}
$L3$ = {c,d,f}
$L4$ = {g,h}

List all words belonging to the following languages:
[1 mark each]

L5 = $(L1 \cup L2) \cap L3 = \{c,d\}$

L6 = $(L1 \cap L2)(L2 \cap L3) = \{cc,cd\}$

L7 = $(\{\varepsilon\}L1\{\varepsilon\}) \cap L3 = \{c\}$

L8 = $(L1 \cup L3)(L3 \cap L4) = (L1 \cup L3)\varnothing = \{\}$

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